

Alberta's Two Maps: What the Audit Found

*A plain-language summary of the 2025–26 Electoral Boundary
Commission forensic audit. vo.20 — April 25, 2026 — [Full monograph](#)*

An Airdrie resident pulls up the new electoral map on her phone. Her street goes to one riding. Two blocks over goes to another. Under the minority commission proposal, her city of 84,000 is cut into four pieces, each quarter attached to a different surrounding region. Nothing about Airdrie changed. Only the map did.

That map is what this audit is about.

The short version

Alberta's Electoral Boundary Commission finished its work on March 23, 2026 and could not agree. The majority of commissioners produced one map; the minority produced a different one. Both are legal. This audit measured both using the same methods, applied identically. Three findings stand out.

1. **The two maps diverge systematically across six structural measures** — population spread, voter concentration, city fragmentation, municipal-boundary anchoring, map shape, and commission-chair-flagged anomalies. None of the six requires election results to calculate.
2. **The direction runs against the places where the governing party is weakest.** Communities most affected by the structural differences in the minority map — Calgary's northwest quadrant, the City of Airdrie, urban areas with established municipal boundaries — are the communities where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.
3. **The process now promoting the minority map departs from Canadian practice.** On April 16, the government set both commission maps aside and assigned redistricting to a five-MLA committee, three from the governing UCP. Most Canadian provinces either require the assembly to consider the commission report first or give commission maps automatic effect unless overridden. Alberta does neither.

The process is its own finding, separate from the maps.

The April 16 decision

The *Electoral Boundaries Commission Act* lets the government refer the commission's recommendations to a legislative committee. On April 16, the government did this. The new committee has five members — three from the governing UCP and one each from the NDP and the Alberta Party. It must report by November 2, 2026 with a 91-seat map.

The referral followed the letter of the Act. It also confirms Alberta's outlier status: most provinces require assembly consideration before referral, and most have commissions chaired by a judge rather than a replaceable public member. The monograph documents three comparator pivots — Quebec 1992, Ontario 1996, British Columbia 2008.

Three things the minority map does differently

SIDEBAR — WHY AIRDRIE MATTERS

Population 84,000. Larger than Red Deer. One council, one tax bill, one school division.

Majority map: 2 divisions. Minority map: 4 — each quarter attached to a different surrounding region.

It splits the City of Airdrie into four pieces. The law caps each electoral division at one-and-a-quarter times the provincial average, so Airdrie needs at least two divisions. The majority map gives it two. The minority map gives it four.

Both maps are legal. The four-way split is a choice.

It ignores municipal boundaries. When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends. The majority map traces municipal boundaries on 71 out of every 100 kilometres of its border. The minority map traces them on only 15 out of every 100. This number has nothing to do with which party benefits — it is measured from publicly available maps, not election results.

One area of Calgary is carved up to concentrate NDP voters into larger-than-average divisions. In Calgary's northwest quadrant, the minority map's divisions average 11.5% above the province-wide population — versus 2.8% on the majority. Concentrating voters into oversized divisions dilutes their weight; redistricting analysts call it *packing*.

Two further structural measures show the same pattern.

What the audit found across five measures

SIDEBAR — WHAT IS A "PRE-REGISTERED PASS"?

Before looking at any results, the audit committed in writing to a list of tests and what would count as passing. When the minority map then *passed* one of those tests (zero packing-cracking neighbour pairs), the result was recorded as a pass. Pre-registration is the strongest defence against cherry-picking — you can't accuse a researcher of fishing if they declared their hypothesis before they had the data.

The table compares the two maps. None of the top six rows uses election results.

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
How evenly distributed the populations are	More even	48% more uneven
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Borders anchored to Statistics Canada census boundaries	80%	17%
Boundaries flagged by the commission chair	0	3
Efficiency gap (an election-fairness measure)	-1.3%	-2.7%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — <i>pre-registered pass</i>

The bottom two rows depend on election results and are less certain. Both efficiency-gap numbers favour the UCP relative to a neutral outcome — expected in Alberta, where concentrated urban NDP support produces a structural conservative tilt under any map. Both fall well below any published gerrymander threshold.

The last row is where the minority map looks *better*: zero packed-cracked adjacency pairs, against three on the majority and three under the current 2019 map — a pre-registered pass (see sidebar).

One way to read the scorecard

Rows of measurements are hard to weigh against each other. One defensible synthesis is a four-category rubric weighted by the principles redistricting law and democratic theory treat as primary.

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SCORECARD — ONE WAY TO READ THE TWO MAPS

A proposed synthesis with explicit weights — not a finding. Other weightings give different grades.

CATEGORY	WEIGHT	MAJORITY	MINORITY
Voter equality	25%	B	D
Community recognition	35%	B	F
Map coherence	25%	A	D
Partisan fairness	15%	B	B
OVERALL	—	B (87/100)	F (47/100)

Why these weights. Community recognition carries the most (35%) because *Reference re Saskatchewan* — the Supreme Court decision defining what redistricting is for — holds that effective representation requires districts that communities can identify as their own. The Act's $\pm 25\%$ population band grounds

voter equality at 25%. Map coherence (25%) reflects the commission's own standard. Partisan fairness is weighted lowest (15%) because both maps fall below any published gerrymander threshold, and the direction of the partisan-fairness finding reverses depending on which measurement method you use.

What produces the minority's F. On community recognition (10/35 under these weights), 14.5% municipal anchoring means 85 out of every 100 kilometres of the minority map's border cuts through territory voters do not recognize from any other civic map. The majority's equivalent is 71%. The Airdrie four-way split compounds the effect — residents must navigate four legislative offices, each primarily identified with somewhere else. Three chair-flagged anomalies — a lasso-shaped corridor, a boundary extension through uninhabited national park land, and a district named for communities together smaller than the community it actually captures — pull the map-coherence score down further.

Where the maps score comparably. Under the lowest-weighted category — partisan fairness — the two maps are effectively tied, and neither shows a serious problem by any published standard.

Was either map a gerrymander?

The honest answer needs two lanes.

LANE 1 — THE PARTISAN-FAIRNESS NUMBERS

Where the two 2026 maps land against the Alberta-derived 4.37% threshold:

SIDEBAR — WHAT IS THE EFFICIENCY GAP?

A wasted-vote count comparing the two parties. The most-cited US redistricting measure, with a 7% threshold researchers use to flag a map. The audit derived four Alberta-calibrated alternatives. The strongest: 4.37%, from running 100,000 different legal Alberta maps through a computer simulation.

MAP	EFFICIENCY GAP (ABSOLUTE)	ABOVE ALBERTA 4.37%?
Majority 2026	1.29%	No — well under
Minority 2026	2.71%	No — under

On Lane 1, neither map is a gerrymander. The minority is closer to the line (about 1.7 percentage points away), but still inside the band a non-partisan computer simulation would produce.

Lane 2 — the structural pattern

Partisan-fairness numbers are not the only way to spot engineering. The audit applied nine further tests of how the map was drawn — all measurable from public maps, none dependent on election results.

TEST	MAJORITY MAP	MINORITY MAP
Border follows existing municipal lines (70–85% Canadian norm)	71% — within norm	15% — 4.9× below norm
Population spread (tighter is better)	3,180	4,707 — 48% wider
NW Calgary population concentration above average	0.4%	12.2%
Boundaries flagged by the commission's own chair	0	3
Airdrie split (constraint minimum: 2)	2 pieces	4 pieces
Minority's published reasons checked symmetrically	n/a	6 of 7 fail
Contested choices with documented public-submission support	n/a	2 have none
Pre-registered structural-irregularity count (4 of 5 = outlier)	0 of 5	5 of 5
Mean-median percentile vs computer simulation	At the 95th-percentile boundary	At the 95th-percentile boundary

On Lane 2, the majority crosses zero structural thresholds. The minority crosses every one of them by a wide margin.

Verdict

	LANE 1: NUMBERS	LANE 2: STRUCTURE
Majority 2026	not a gerrymander	not a gerrymander
Minority 2026	not a gerrymander	exhibits a gerrymander signature

OPINION — THE EFFECTS, REPORTED PLAINLY.

The measurement work returns a split result: the minority sits below threshold on the partisan-fairness numbers and above threshold on the structural pattern. Each individual structural signal has an innocent explanation available; the cumulative pattern is what the academic redistricting literature treats as a gerrymander signature. Whether the pattern's most parsimonious explanation is engineering, unlucky drafting, or some combination is the inference the reader is invited to weigh. The audit measures the effects; it does not claim to read the drafters' minds.

BOTTOM LINE:

below threshold on the numbers; above threshold on the structure. The literature weights the structural pattern more heavily; the monograph (§6.2) sets out both readings without picking one.

How close did the minority map come?

- **On the partisan-fairness numbers:** about 1.7 percentage points under the Alberta-calibrated 4.37% threshold. Closer to the line than the majority, but still under.
- **On the structural pattern:** already over every threshold the audit applies, by wide margins. The 4.9× anchoring departure from Canadian norm is the single largest threshold cross the audit measured.

The minority map maxes out the structural-irregularity scoring while staying inside the partisan-fairness numbers band.

SCORECARD — RETRACTION CONDITIONS

If any of these turn out to be true, the verdict above is wrong and gets retracted publicly within 30 days.

1. **A counter-map exists.** *Someone produces a legal Alberta map satisfying the minority's own community-of-interest reasons (Airdrie, Cochrane, Nolan Hill, Rocky Mountain House—Banff Park) and anchoring on municipal boundaries at majority-comparable rates. Open challenge — Issue #14 on the audit's GitHub repository.*
 2. **The Lundy committee map looks like the minority.** *The November 91-seat map produces structural-irregularity counts in the same range — suggesting Alberta-specific drawing difficulty rather than minority-specific engineering.*
 3. **A pre-2026 internal commission document surfaces.** *Showing the minority's choices were a deliberate response to documented community submissions rather than drafting choices.*
 4. **The cross-election reversal extends to recent polling.** *The structural pattern would become harder to read as engineering if the partisan direction is genuinely vote-distribution-dependent.*
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What this audit is now for

In policy terms, the verdicts above are moot. On April 16 the government set both commission maps aside; Alberta's 2027 map will come from the Lundy committee. Neither commission map is a candidate to become law.

The work itself is not moot. The commission audit calibrated Alberta-specific thresholds — structural-irregularity tests, anchoring comparators, a 100,000-plan computer ensemble — against three real Alberta maps (the 2019 enacted boundaries plus the two 2026 commission proposals, applied symmetrically). That framework will be re-pointed at the Lundy committee's map the day it is published, reported by the same standard you have just read.

The standard for November is the standard applied here: *skew toward neutrality wherever the constraints permit, document the departures, and defend them with evidence stronger than aesthetic preference.* The same scripts will answer, publicly, against the same thresholds — because we'll see if they don't.

SIDEBAR — TWO REMINDERS FOR THE NOVEMBER AUDIT

Alberta's geography will always favour conservatives. NDP voters concentrate in city cores; UCP voters spread across efficient-margin rural ridings. A neutrally-drawn Alberta map produces a UCP-favourable efficiency gap as a starting point — that's geography, not gerrymandering.

Many things count as gerrymandering, and not all show up in the partisan-fairness numbers. The minority map sat sub-threshold on every Alberta-calibrated EG line while crossing every structural threshold by a

What the audit does not claim

The audit does not claim the minority map was drawn to help the UCP win seats — intent cannot be proven from maps alone. It claims only that the differences are measurable, consistent, and not explained by Alberta's geography.

Both maps satisfy the *Electoral Boundaries Commission Act*.

The partisan-fairness numbers are sensitive to which election's results you use. Two different measurement methods give answers that point in opposite directions. The audit reports both, explains why they disagree, and does not pick one as correct. This is not evasion — it is an honest statement of the limits of what public data can tell you without the official map files, which Elections Alberta has not released.

What you can do

1. **Read the full findings.** The monograph ([report_academic.md](#)) contains every number, every source, and every caveat. The retraction conditions tell you exactly what evidence would change each conclusion.
 2. **Ask your MLA what the committee's process will be.** - Will the committee publish the evaluation criteria it uses *before* drawing a single line? - Will it release the range of maps it considered, not just the one it chose? - Will it publish any prompts or inputs given to AI tools?
 3. **Request the official shapefiles.** Elections Alberta has not published the 2026 electoral division map files. Without these, some of this audit's calculations cannot be fully verified or corrected. A FOIP request would require their release.
 4. **Share this report.** The full monograph is written for researchers and journalists. This report is written for everyone else. If you think the commission's process matters, share what you found here.
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About this audit

This audit was produced by Will Conner, a computer science student at Mount Royal University in Calgary. It was conducted as an independent research project with no funding from any political party or advocacy organization.

The audit used publicly available data: Elections Alberta's 2023 Statement of Vote, Statistics Canada's 2021 Census, and the commission's 2026 final report. All code is published in this repository and can be run independently to reproduce every number.

An AI assistant (Claude, made by Anthropic) helped draft and check the text. All factual claims were verified by the author against primary sources. The audit declares this openly because the same standard it applies to the commission — transparency about process — applies to the audit itself.

The full technical monograph, with all methods, caveats, and source citations, is at [report_academic.md](#).